Cluster 03 – Data-driven decision making

Shared themes in the middle, pairs on the edges, each topic outside · Elena · Giovanni · David

IN PLAIN WORDS

Someone chooses between options. Data predicts what each option does before it exists; the model decides which result counts as better. **“What makes it correct is a model decision.”** (Iacopo, RM_L04)

The average vs the single case
- personas from labels → each person
- a table by district → each tree position

Comfort in public space
- how one person feels in a place
- shade to walk in, a dry street
- thermal comfort outdoors

Predict before building
- photo ratings
- shade proxy + runoff model
- UTCI simulation

A tool for those who decide
- web app for sensory needs
- Grasshopper + web app
- QGIS + Python workflow

DATA-DRIVEN DECISION MAKING

what counts as “better”?

Check against reality
- participants’ ratings
- measured rain interception
- measured air temp., humidity, wind

What is left out?
- senses a photo cannot carry
- roots, tree pit, hot nights
- one value per scenario

Beat the simple rule
- a group average
- the city’s replacement rule
- today’s conditions

“Cooler” ≠ “more comfortable”
- perceived comfort
- thermal comfort

back to what people live

calibrate the proxy

if the simple rule wins, it is still a result

ELENA

How one person will feel in a public space, from a photo

starts from: the optimal against the individual

Empirical gap no test of whether a persona reflects one person’s senses in public space

GIOVANNI

Which tree replaces which tree on a Barcelona street

starts from: the optimal against the rules

Methodological gap no method picks species per tree position on heat and runoff together, inside a city’s replacement rules

Heat measured as UTCI
- Ladybug UTCI for each tree layout
- UTCI for vegetation vs water scenarios

Greening inside existing constraints
- street trees in existing pits
- native vegetation in existing spaces

Water as a factor
- runoff from a storm
- water as a cooling element

DAVID

Native vegetation and water for outdoor thermal comfort

starts from: ? (David)

Gap limited evidence on when combining native vegetation and water improves pedestrian thermal comfort more than either alone

EVERY BRANCH MAKES THE SAME FIVE MODEL DECISIONS (ELENA; IACOPO; MATHILDE: “MAKE EACH SIMPLIFICATION EXPLICIT, WITH ITS RISK”)

Data	Sampling	Resolution	Representation	Model
what, at what scale	which people, places, events	one person, 1 m, one hour	a lived quality becomes a number	what counts as correct